39) Find the dual of each of these compound propositions

a) $p \land \neg q \land \neg r$ Dual: $p \lor \neg q \lor \neg r$

b) $(p \land q \land r) \lor s$ Dual: $(p \lor q \lor r) \land s$

c) $(p \lor F) \land (q \lor T)$ Dual: $(p \land T) \lor (q \land F)$

41) Show that $(s^*)^* = s$ when s is a compound proposition.

$\boxed{\text{Soln:}}$ The proof will be by strong mathematical induction on n, which is the number of $\land$ and $\lor$ present in a compound proposition s.

$\boxed{\text{Base Case:}}$ $n=0$. $\boxed{\therefore s \text{ is an atomic proposition by defn.}}$

The truth value of s is either True or False. Let, wlog it be True.

$\therefore (s^*)$ has a truth value of False (by defn. of dual)

$\therefore (s^*)^*$ has a truth value of True (by defn. of dual)

So $(s^*)^* = s$ in this situation. We can prove similarly that $(s^*)^* = s$ when the truth value of s is False.

$\boxed{\text{Induction Hypothesis:}}$ $n=K$. For all compound propositions s having no. of logical operators ($\land$ or $\lor$) $\leq K$, $(s^*)^* = s$.

$\boxed{\text{Induction Step:}}$ $n=K+1$. The compound proposition s can be written in one of the following ways ① $s = p \land q$ OR ② $s = p \lor q$, where both p, q has $\leq K$ propositional logical operators ($\land$ or $\lor$), o.w. s would have more than $(K+1)$ logical operators.

$\boxed{\text{Case 1:}}$ $s = p \land q$. $\therefore s^* = p^* \lor q^*$ (Follows from the defn. of dual)

$(s^*)^* = (p^*)^* \land (q^*)^*$ (Follows from the defn. of dual)

$= p \land q$ (From Induction Hypothesis)

$= s$

$\boxed{\text{Case 2:}}$ $s = p \lor q$. $\therefore s^* = p^* \land q^*$ (Follows from the defn. of dual)

$(s^*)^* = (p^*)^* \lor (q^*)^*$ (Follows from the defn. of dual)

$= p \lor q$ (From Induction Hypothesis)

$= s$

$\boxed{\therefore (s^*)^* = s}$ □